bpfilter: a BPF-based packet filtering framework

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- Software Engineer @ Meta, member of the Linux Userspace team.
  - Contributing and promotion userspace open source tools at Meta.
- Working on a BPF-based packet filtering framework.
- This talk is about firewall.

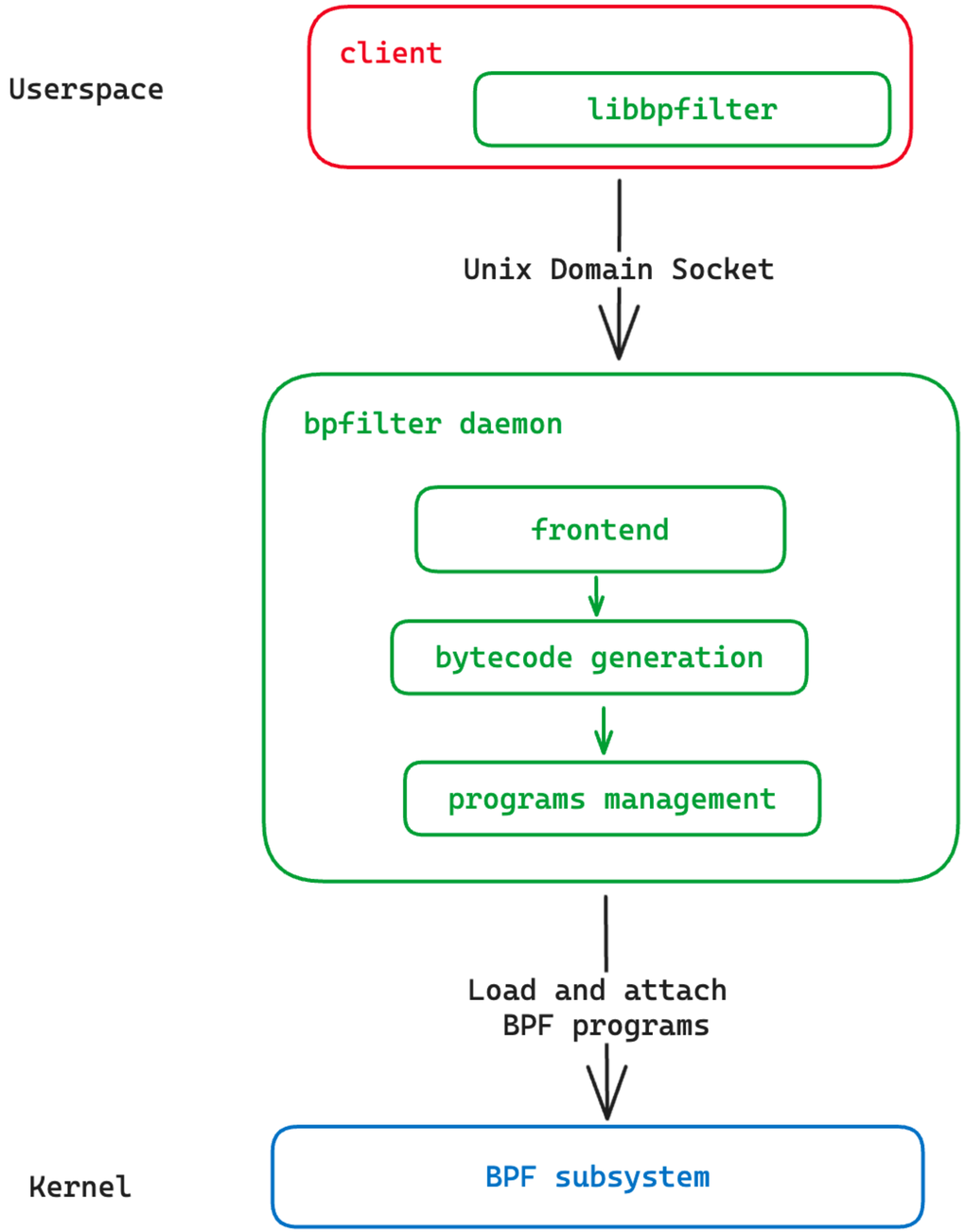
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BPFILTER ORIGINS

- Originally developed as a UMH by Alexei Starovoitov, Daniel Borkmann, and David Miller.
- Catches `iptables-legacy` calls to `getsockopt()` and `setsockopt()`.
- Translates filtering rules into XDP BPF programs.
- Abandoned around 2021.
- Developed as a userspace daemon since earlier this year.

bpfilter structure

- Client is linked to `libbpfilter`.
- Communication between the library and the daemon goes through a Unix Domain Socket.
- `bpfilter` daemon will translate the client-specific data into a generic format, and generate the BPF bytecode.
- The daemon manages the BPF programs' lifetime.



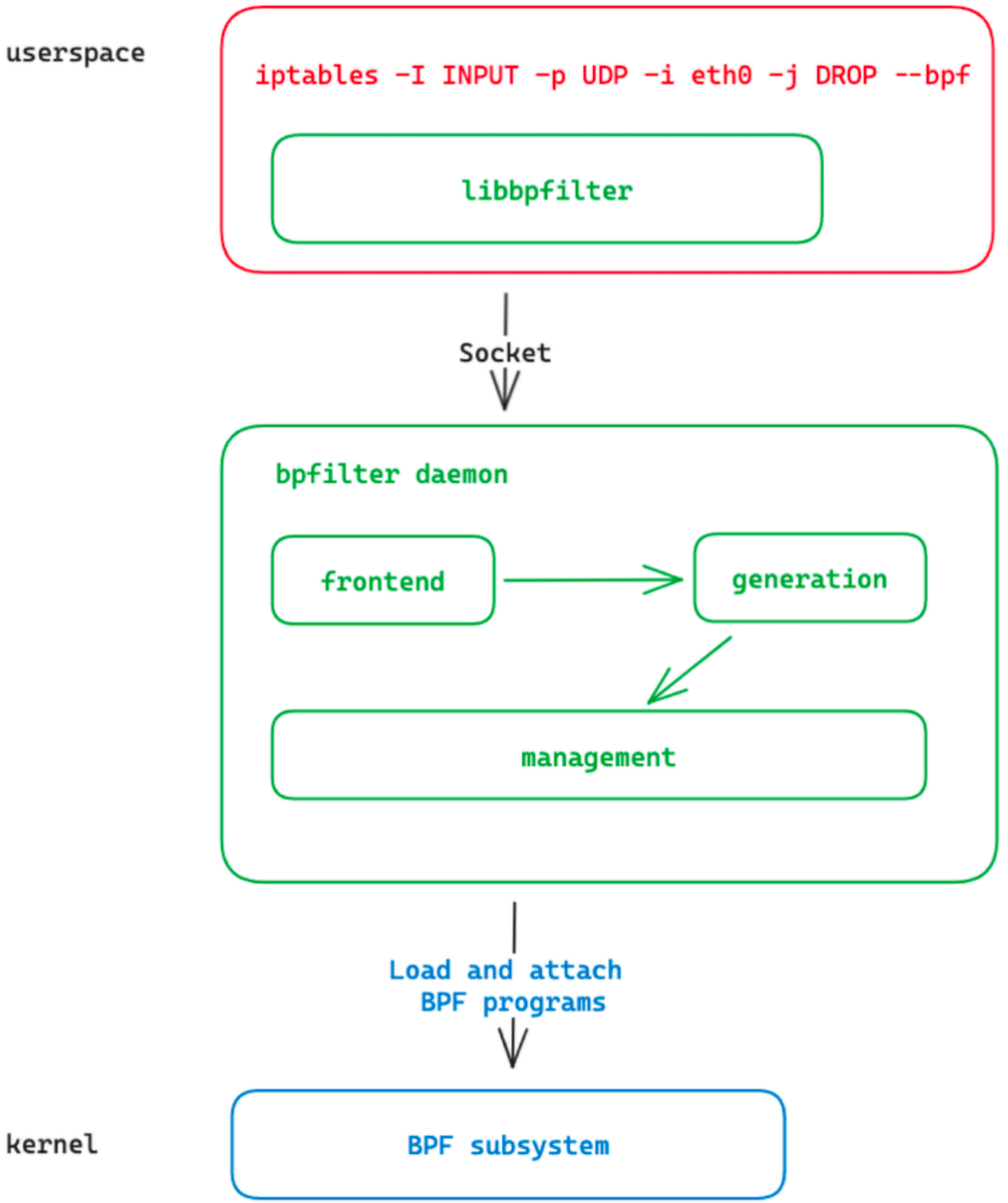
libbpfilter

- Linked to the client, uses its data structures.
- Provides a lightweight interface to the daemon.
- Client-specific data is translated by the daemon.

```
-         ret = setsockopt(handle->sockfd, TC_IPPROTO, SO_SET_REPLACE, repl,
-                          sizeof(*repl) + repl->size);
+ #if defined(ENABLE_BPFILTER) && !defined(_LIBIP6TC_H)
+     if (handle->use_bpf)
+         ret = bf_ipt_replace(repl);
+     else
+ #endif
+     ret = setsockopt(handle->sockfd, TC_IPPROTO, SO_SET_REPLACE, repl,
+                      sizeof(*repl) + repl->size);
```

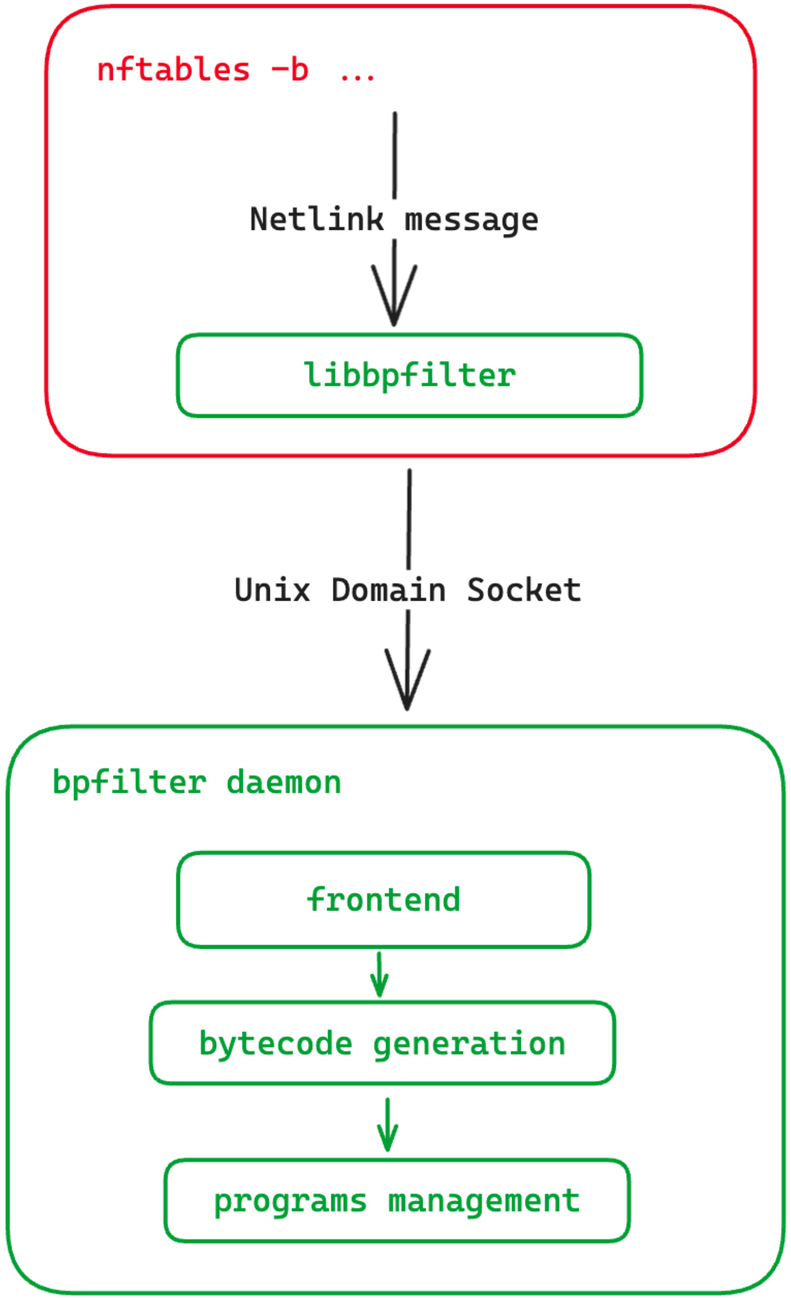
iptables and bpfilter

- iptables read/write the whole ruleset from/to bpfilter through the Unix Domain Socket.
- Programs are fully re-generated on rule change.
- Uses BPF_NETFILTER hooks (6.4).
- Support counters, IP, port, and protocol filtering.



nftables and bpfilter

- nftables sends Netlink messages directly to bpfilter through the Unix Domain Socket.
- Parsing is performed in the daemon using libnl.
- Uses XDP hook to offload nftables rules.



bpfilter: BPF bytecode generation

- Write BPF instructions directly into a memory buffer.
- Fixups to modify the bytecode once the generation is complete.
- Support for custom functions and stubs.
- Support XDP, TC, and BPF_NETFILTER program types.
- Uses BPF dynamic pointers to access packet data.
- Per-rule-interface packets and bytes counters.

```
// BF_ARG_1: counters map file descriptor.
EMIT_FIXUP(program, BF_CODEGEN_FIXUP_MAP_FD, BPF_MOV64_IMM(BF_ARG_1, 0));

// BF_ARG_2: index of the current rule in counters map.
EMIT(program, BPF_MOV32_IMM(BF_ARG_2, bf_list_size(rules)));

// BF_ARG_3: packet size, from the context.
EMIT(program,
     BPF_LDX_MEM(BPF_DW, BF_ARG_3, BF_REG_CTX, BF_PROG_CTX_OFF(pkt_size)));

EMIT_FIXUP_CALL(program, BF_CODEGEN_FIXUP_FUNCTION_ADD_COUNTER);

EMIT(program,
     BPF_MOV64_IMM(BF_REG_RET, program->runtime.ops->get_verdict(policy)));
EMIT(program, BPF_EXIT_INSN());
```

Bytecode generation

```
24: (b7) r1 = 6
25: (b4) w2 = 0
26: (79) r3 = *(u64 *)(r9 +24)
27: (85) call pc+2#bpf_prog_2b9b10dee25f87b3_F
28: (b7) r0 = 2
29: (95) exit
```

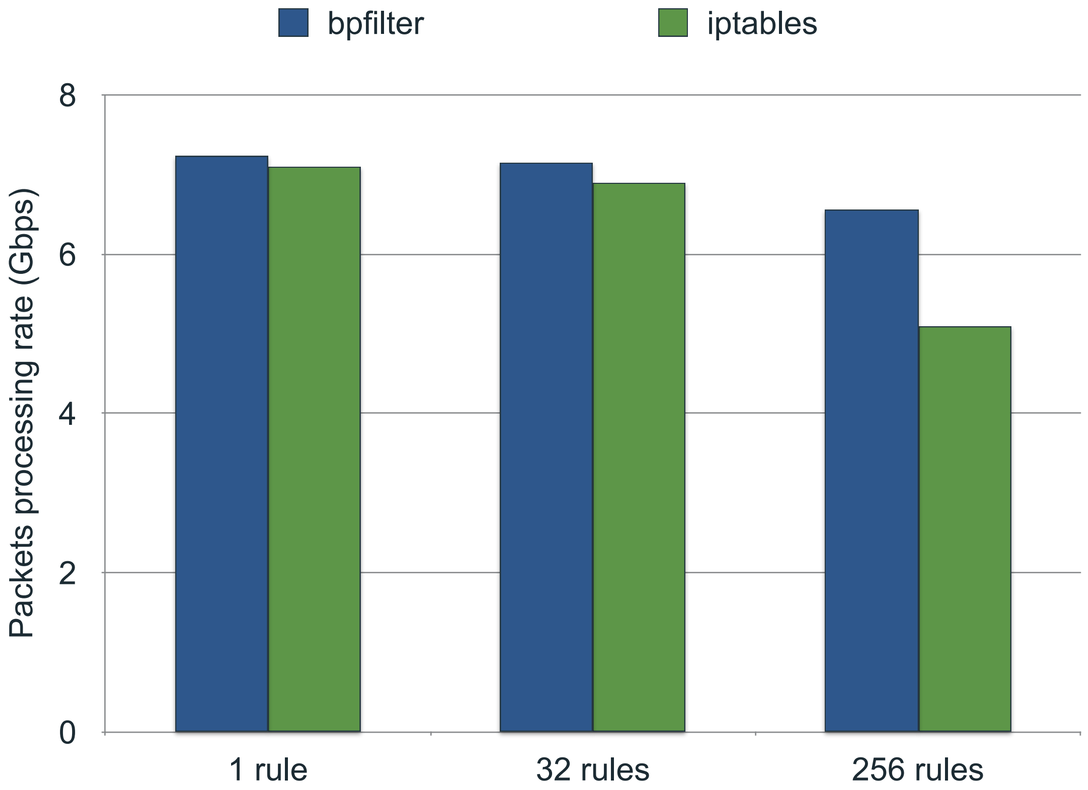
`bpftool`'s output

USAGE

```
› sudo iptables -I INPUT -j DROP -p UDP -i enp0s5 --bpf
› sudo bpftool prog list
75: netfilter  name bpfltr_02000002  tag 43bf0d20e76f4997  gpl
        loaded_at 2023-09-12T11:51:28+0200  uid 0
        xlated 832B  jited 752B  memlock 4096B  map_ids 25
[...]
86: netfilter  name bpfltr_04000003  tag caf36bf4fc15b318  gpl
        loaded_at 2023-09-12T11:51:34+0200  uid 0
        xlated 832B  jited 752B  memlock 4096B  map_ids 36
› sudo ls /sys/fs/bpf
bpfltr_m_02000002  bpfltr_m_03000002  bpfltr_m_04000002  bpfltr_p_02000002  bpfltr_p_03000002  bpfltr_p_04000002
bpfltr_m_02000003  bpfltr_m_03000003  bpfltr_m_04000003  bpfltr_p_02000003  bpfltr_p_03000003  bpfltr_p_04000003
```

Performance

- Dropping incoming packets on `bpfilter` and `iptables`.
- What about `nftables`?



Resources

- `bpfilter` repository: github.com/facebook/bpfilter
- `iptables` PoC: github.com/qdeslandes/iptables/tree/bpfilter
- Nftables PoC: github.com/qdeslandes/nftables


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